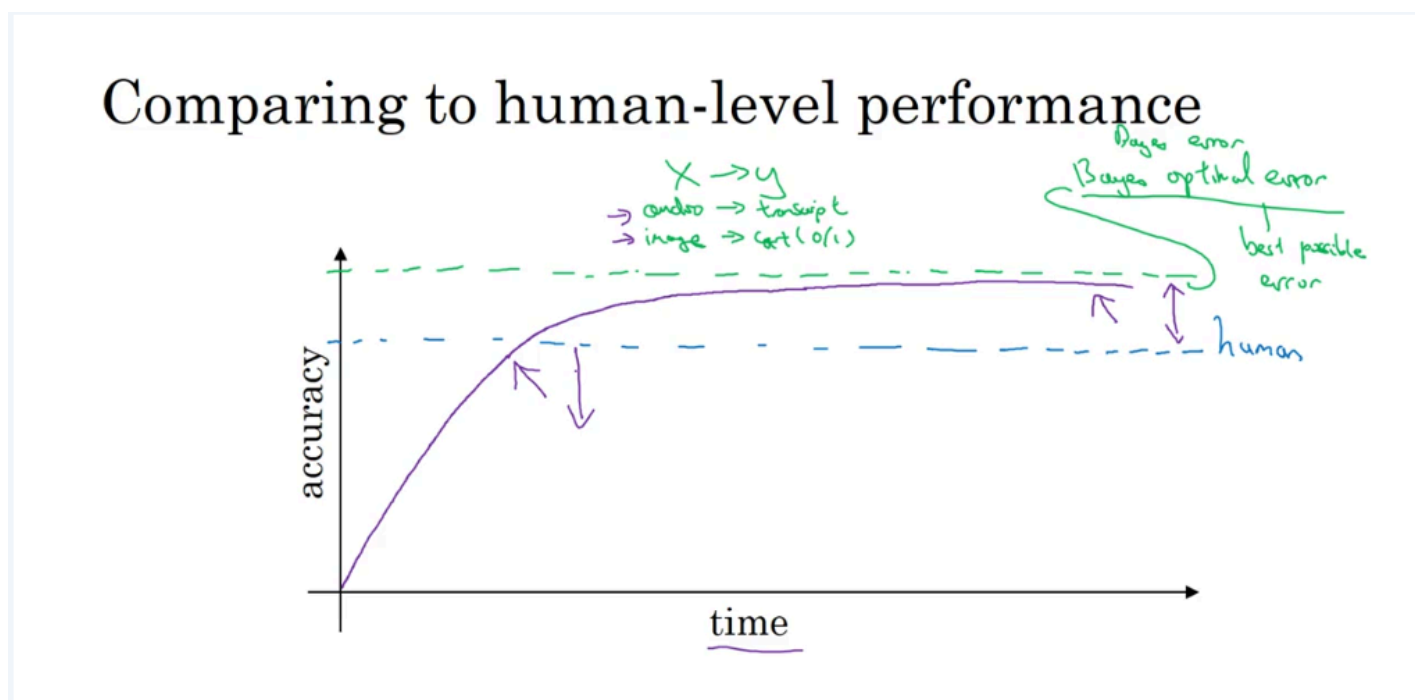


Comparing to Human-level Performance

Why human-level performance?



Understanding Human-Level Performance

- Advances in deep learning have made it feasible for machine learning algorithms to compete with human performance in various applications.
- The workflow for designing machine learning systems becomes more efficient when aiming to mimic human capabilities.

Progress in Machine Learning

- Progress tends to be rapid as machine learning approaches human-level performance, but may slow down once it surpasses it.
- The concept of [Bayes optimal error](#) represents the theoretical best performance that can be achieved, which cannot be surpassed regardless of advancements.

Challenges Beyond Human-Level Performance

- Once machine learning algorithms exceed human performance, the potential for further improvement may be limited due to the proximity to Bayes optimal error.
- Certain improvement strategies, such as obtaining labeled data from humans, become less effective when the algorithm performs better than humans, making it harder to enhance performance further.

Why compare to human-level performance

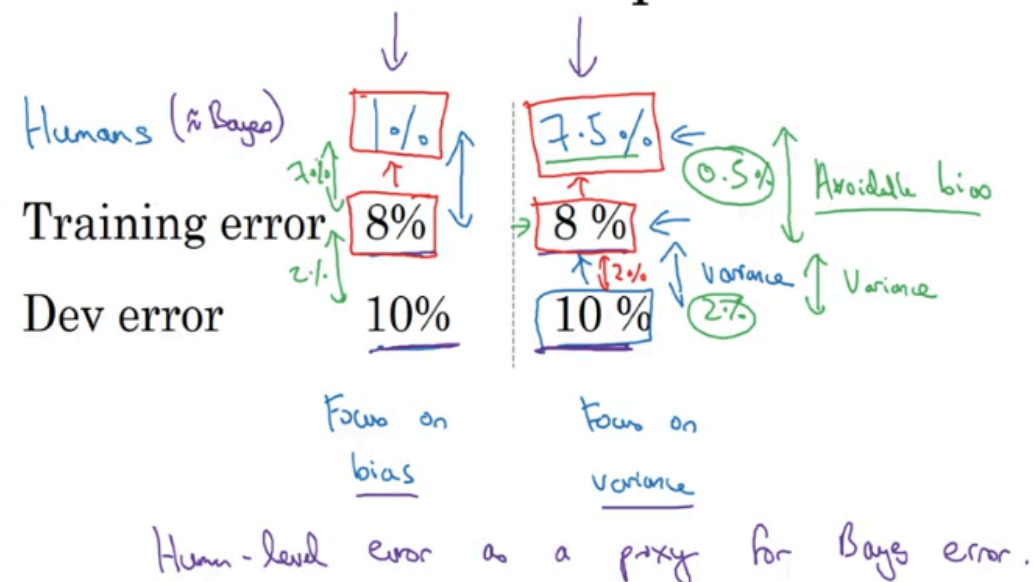
Humans are quite good at a lot of tasks. So long as ML is worse than humans, you can:

- Get labeled data from humans. (x, y)
- Gain insight from manual error analysis:
Why did a person get this right?
- Better analysis of bias/variance.

Avoidable Bias

the term avoidable bias acknowledges that there's some bias or some minimum level of error that you just cannot get below

Cat classification example



Understanding Human-Level Performance

- Human-level performance serves as a benchmark for evaluating machine learning algorithms, indicating how well they should perform.
- If human-level error is low (e.g., 1%), a higher training error (e.g., 8%) suggests the algorithm is not fitting the training data well.

Bias vs. Variance

- In cases where human-level performance is high, focus on reducing bias by improving the algorithm's training performance.
- Conversely, if human-level performance is already close to the algorithm's performance, the focus should shift to reducing variance to avoid overfitting.

Avoidable Bias and Variance

- The difference between the training error and the estimated Bayes error is termed "avoidable bias," which indicates the potential for improvement.
- The gap between training error and development error reflects the variance problem, where reducing this gap is crucial for better model performance.

Overall, understanding human-level performance helps in deciding whether to prioritize bias reduction or variance reduction strategies in machine learning projects.

Understanding Human-level Performance

Human-level error as a proxy for Bayes error

Medical image classification example:



Suppose:

- (a) Typical human 3 % error
- (b) Typical doctor 1 % error
- (c) Experienced doctor 0.7 % error
- (d) Team of experienced doctors .. 0.5 % error

What is “human-level” error?

Andrew Ng

Is human-level error 3%, 1%, 0.7% or 0.5%?

I would urge you to bear in mind that one of the most useful ways to think of human error is as a proxy or an estimate for Bayes error.

But here's how I would define human-level error. Which is if you want a proxy or an estimate for Bayes error, then given that a team of experienced doctors discussing and debating can achieve 0.5% error, we know that Bayes error is less than equal to 0.5%.

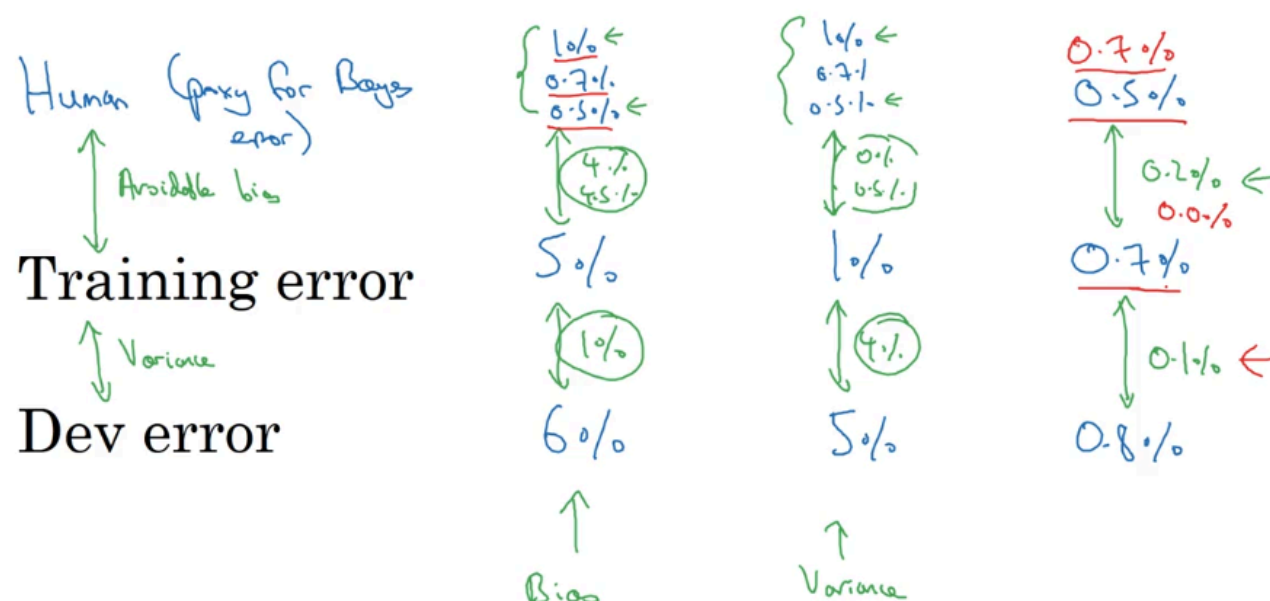
So because some system, team of these doctors can achieve 0.5% error, so by definition, this directly, optimal error has got to be 0.5% or lower. We don't know how much better it is, maybe there's a even larger team of even more experienced doctors who could do even better, so maybe it's even a little bit better than 0.5%. But we know the optimal error cannot be higher than 0.5%.

So what I would do in this setting is use 0.5% as our estimate for Bayes error. So I would define human-level performance as 0.5%

So maybe the takeaway from this is to be clear about what your purpose is in defining the term human-level error. And if it is to show that you can surpass a single human and therefore argue for deploying your system in some context, maybe this is the appropriate definition.

Error analysis example

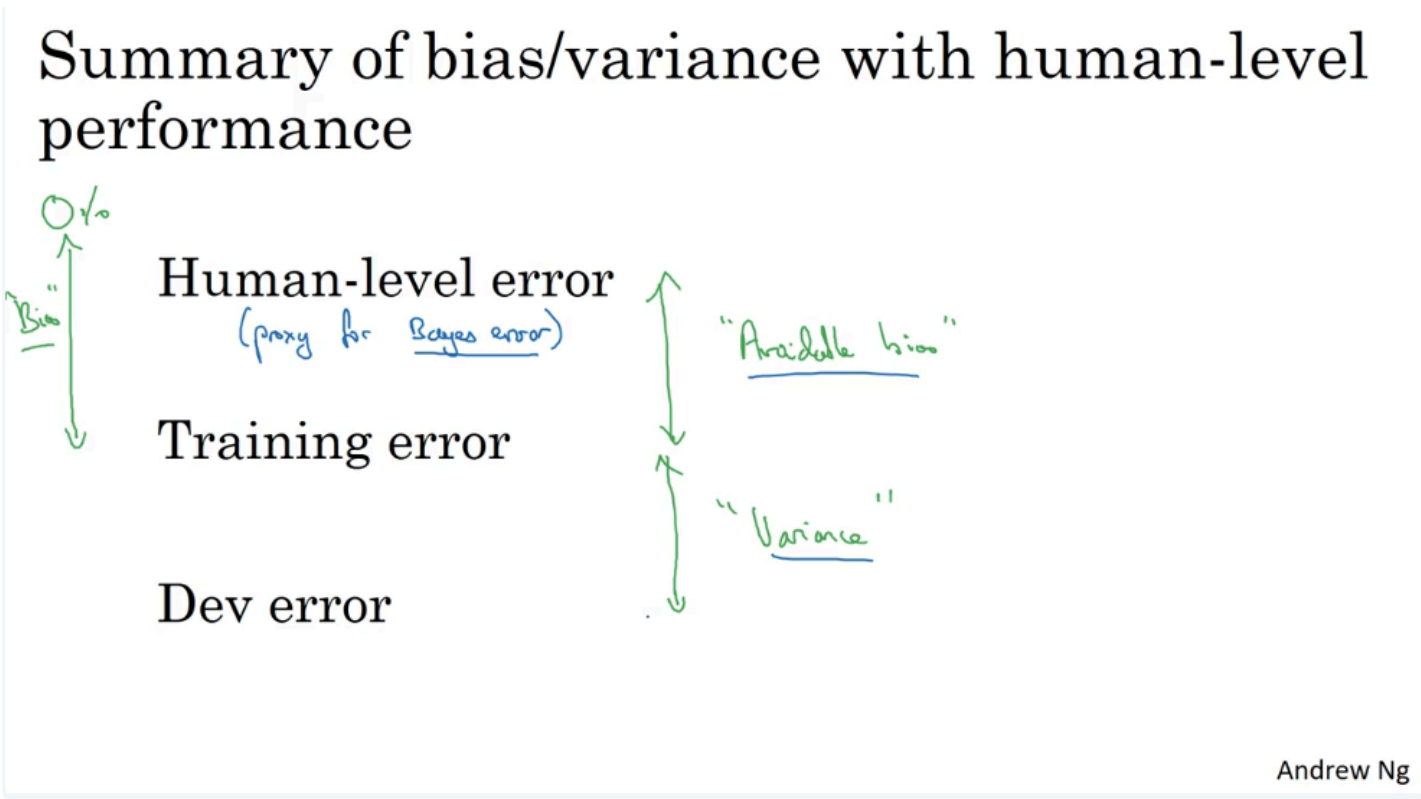
Error analysis example



Andrew Ng

- If training error is much higher than human 5% (4-4.5% in difference) and the dev error is 6% (1% in difference) this could be bias → try using bigger network

- If training error isn't different much compared to human error, but the dev error is much higher (4% in diffence) this could be variance → try using regularization or finding a bigger training set

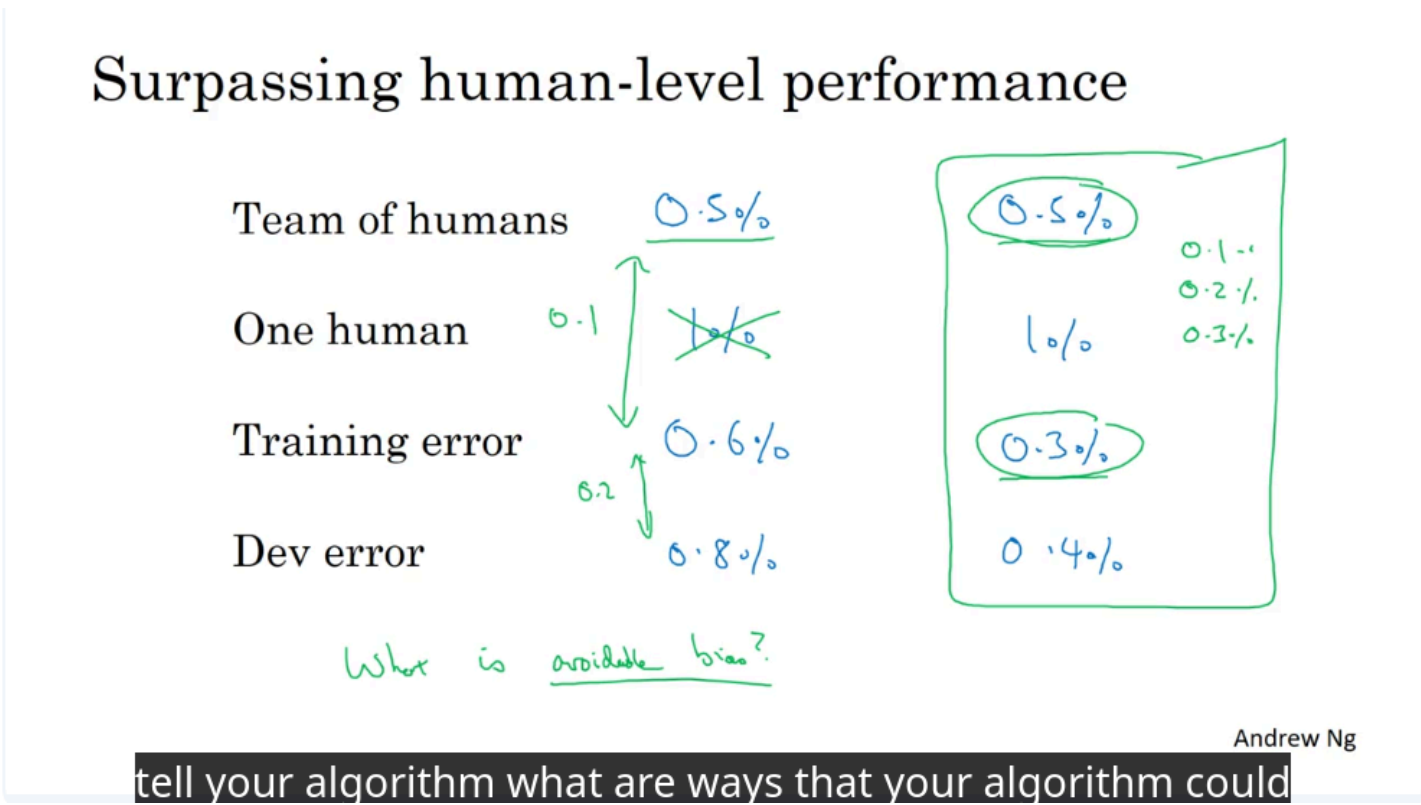


therefore make better decisions on whether to focus on bias reduction tactics, or on variance reduction tactics.

So to recap, having an estimate of human-level performance gives you an estimate of Bayes error. And this allows you to more quickly make decisions as to whether you should focus on trying to reduce a bias or trying to reduce the variance of your algorithm. And these techniques will tend to work well until you surpass human-level performance, whereupon you might no longer have a good estimate of Bayes error that still helps you make this decision really clearly.

Surpassing Human-level Performance

Surpass: Vượt trội



Problems where ML significantly surpasses human-level performance

- - Online advertising
- - Product recommendations
- - Logistics (predicting transit time)
- - Loan approvals

Structural data
Not natural perception
Lots of data

Andrew Ng

Today, models can also good at these tasks:

- Speech recognition
- Some Computer vision task
- Medical
 - Read ECGs
 - Skin cancer diagnosis

Improving your Model Performance

The two fundamental assumptions of supervised learning

1. You can fit the training set pretty well. 
~ Avoidable bias
2. The training set performance generalizes pretty well to the dev/test set. 
~ Variance



such as regularization or getting more training data.

Andrew Ng

Reducing (avoidable) bias and variance

